Activity 9 Pre-Lab — EAS Bromination of Aniline / N-Phenylacetamide

Organic Lab · electrophilic aromatic substitution

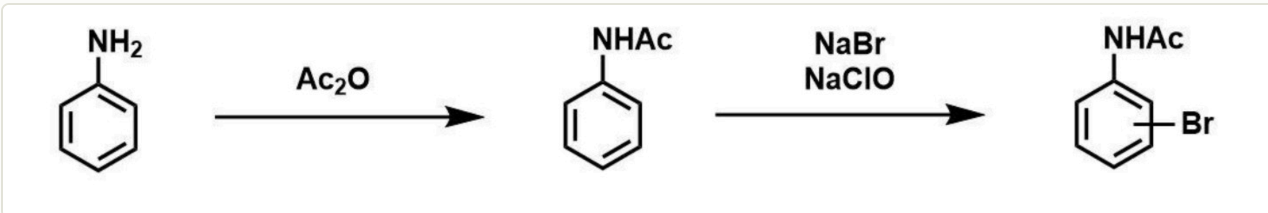
[Your name] & [partner]
[MM/DD/YY]

Black = write this in your notebook **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Study **electrophilic aromatic substitution (EAS)**: first **protect** aniline's amine as an amide (N-phenylacetamide), then **brominate** the ring. This shows how an –NHR group is a strong **o/p-activator** and how protection tempers its reactivity so you get clean **para** product.

✍ **DRAW — reproduce this reaction in your notebook (from the procedure)**



Step 1 protects the amine as the amide (Ac_2O); Step 2 brominates — Br^+ is made in situ from NaBr + NaOCl (bleach) in acetic acid; the amide is an o/p-director so you get the **para** product.

Chemical Reagents & Safety Table

CONTEXT › Required (8): aniline, acetic anhydride, sodium hypochlorite, 95% ethanol, glacial acetic acid, sodium bromide, sodium thiosulfate, sodium hydroxide. Typical SDS values — cross-check the SDS tile. **Aniline is quite toxic** — treat carefully.

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
Aniline	Starting material (limiting)	93.13	~1 mL (0.5 mL ×2, ~0.011 mol)	Colorless–brown oily liquid; bp 184 °C; d 1.02 g/mL	toxic, corrosive, health, environ.
Acetic anhydride	Acetylating (protecting) agent	102.09	2 mL	Colorless liquid; bp 139 °C; d 1.08 g/mL	flammable, corrosive, irritant
Sodium hypochlorite (bleach)	Oxidant → generates Br^+	74.44	18 mL	Pale greenish aqueous liquid	corrosive, environ.
Sodium bromide	Bromine source	102.89	~2.8 g	White crystalline solid	irritant
Glacial acetic acid	Solvent / acid medium	60.05	10 mL	Colorless liquid; bp 118 °C; d 1.05 g/mL	flammable, corrosive
95% Ethanol	Recrystallization solvent	46.07	10 mL	Colorless liquid; bp 78 °C; d 0.79 g/mL	flammable, irritant
Sodium thiosulfate	Quenches excess oxidant	158.11	as directed	White solid (aq. soln)	— low hazard
Sodium hydroxide (0.5 M)	Dissolves/works up amide	40.00	3 × 10 mL	Colorless aqueous solution	corrosive

GRAB FIRST: 50-mL round-bottom + stir bar · stir plate + clamp · ice-water bath · 2× 1-mL syringes (aniline) + 25-mL Erlenmeyer + 400-mL beaker · latex gloves (aniline) then nitrile · steam bath · 125-mL Erlenmeyer · Büchner funnel + filter flask + vacuum · graduated cylinders.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Protect: aniline → acetanilide

1. Instructor adds **2 mL acetic anhydride** to a clamped **50-mL RBF** + stir bar; stopper; lower into an **ice bath** on a stir plate.
2. At the aniline hood (**latex gloves**), draw **0.5 mL aniline into each of 2 syringes** (record exact volume for theoretical yield). Aniline is toxic + absorbs through skin — don't overdraw the syringe.
3. Add the aniline **dropwise over 5 min** with stirring → mixture solidifies (acetanilide); break up lumps with a spatula. Clean syringes with acetone into aniline waste; switch to **nitrile gloves**.
4. Add **10 mL 0.5 M NaOH**; warm on steam bath ~1 min to loosen solid; pour into a 125-mL Erlenmeyer; rinse RBF with **2 more × 10 mL NaOH** into the same flask.
5. Cool to RT then in ice ~10 min; collect the colorless **N-phenylacetamide** by **vacuum filtration**; save an IR-size sample + take IR.

Part 2 — Brominate

6. Dissolve the acetanilide in **10 mL glacial acetic acid + 10 mL 95% ethanol**; add **~2.8 g NaBr**, stir to dissolve.
7. Measure **18 mL bleach (NaOCl)** and add to the reaction; stir. Bleach + NaBr in acid makes the Br^+ electrophile in situ. Never mix bleach with straight acid elsewhere — toxic gas.
8. Quench excess oxidant with **sodium thiosulfate**, precipitate, **vacuum filter**, and **recrystallize from 95% ethanol**; weigh → % yield; mp / IR / NMR (para pattern).

✍ **WRITE**
aniline is limiting → theoretical acetanilide → theoretical 4-bromoacetanilide; % yield each step.

Experiment Notes (in lab)

Exact aniline volume: ______ mL

Part 1 — solidifying / color:

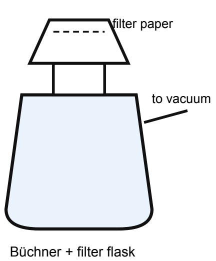
Acetanilide mass + IR notes:

Part 2 — on adding bleach (color/heat):

Product mass / mp: ______

% yield · para evidence (NMR):

Setup — draw this



vacuum filtration

CONTEXT (don't copy) ▸
Why protect the amine first? free aniline is SO activated it over-brominates (tri-bromo) and its lone pair also reacts with the oxidant. Acetylating to the amide dials the activation down to a controllable **o/p-director**, so you get mono-**para** product.

Where does Br^+ come from? NaOCl (bleach) oxidizes Br^- (from NaBr) in acetic acid to an electrophilic bromine species — greener than using liquid Br_2.

Clean-up & Conclusions

- Aniline rinses → labeled aniline waste; latex gloves → solid waste; filtrates → aqueous waste; organic solvents → organic waste; keep bleach away from acids; bench task.
- **Conclusions** (~4 sentences after lab): % yields; that the –NHC(O)CH_3 group directed bromination **para** (NMR: two doublets, para pattern); why protecting the amine was necessary; a loss source.